Computer Forensics Case Studies

Case Study 4 Investigating Rhion Possession Evidence

Student Assessment Brief

SBTDF204_CaseStudy4_**2025/FWSD/11478**

HUSSEINI SULEIMAN

2025/FWSD/11478

fwsd2511478@students.icdfa.edu.ng

DATE: 26Th October, 2026

Assessment detail	Information
Course	SBT-DF204 Computer Forensics Case Studies
Assessment	Case Study 4 Individual recovered-media and network-traffic investigation
Case title	Investigating Rhion Possession Evidence
Submission format	Written report with labelled screenshots, timeline, and evidence log
Deadline and maximum score	As published in E-Campus

This individual case study brings together disk-image recovery, steganography, FTP, HTTP, and Wireshark analysis. You will work from the authorised course evidence package and produce an evidence-led account of what the artefacts do and do not establish. Your report must clearly separate direct observations from inferences about file transfer, concealment, possession, or the person who used a device.

Case scenario

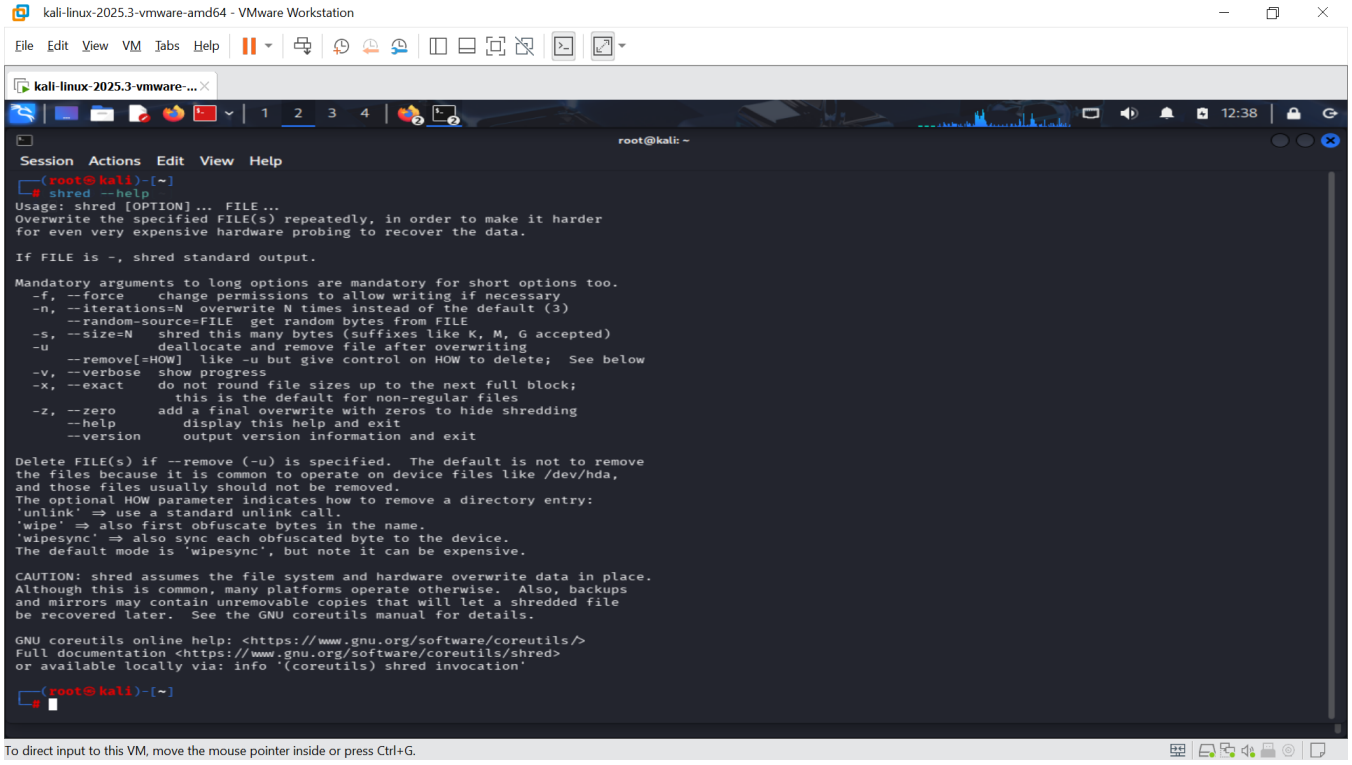
This is a fictional training investigation concerning imagery that may meet an illegal-possession threshold in the case scenario. The evidence package includes a seized USB disk image and network traces associated with FTP and HTTP activity. Investigators need to know whether deleted files, hidden data, reconstructed network objects, and the timing of recorded transfers form a defensible sequence. The task is an offline examination of supplied training artefacts; it does not authorise any contact with people, accounts, systems, hosts, or services named in the material.

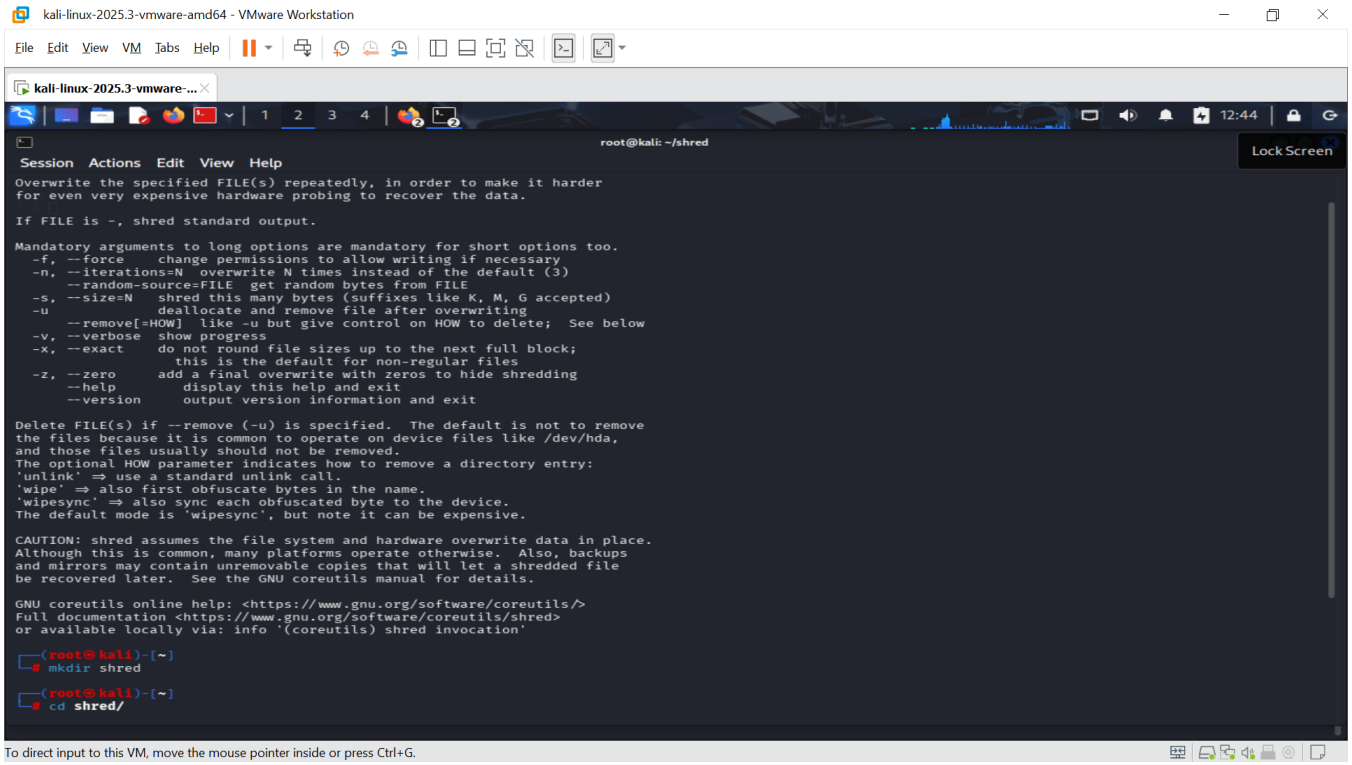
Case materials

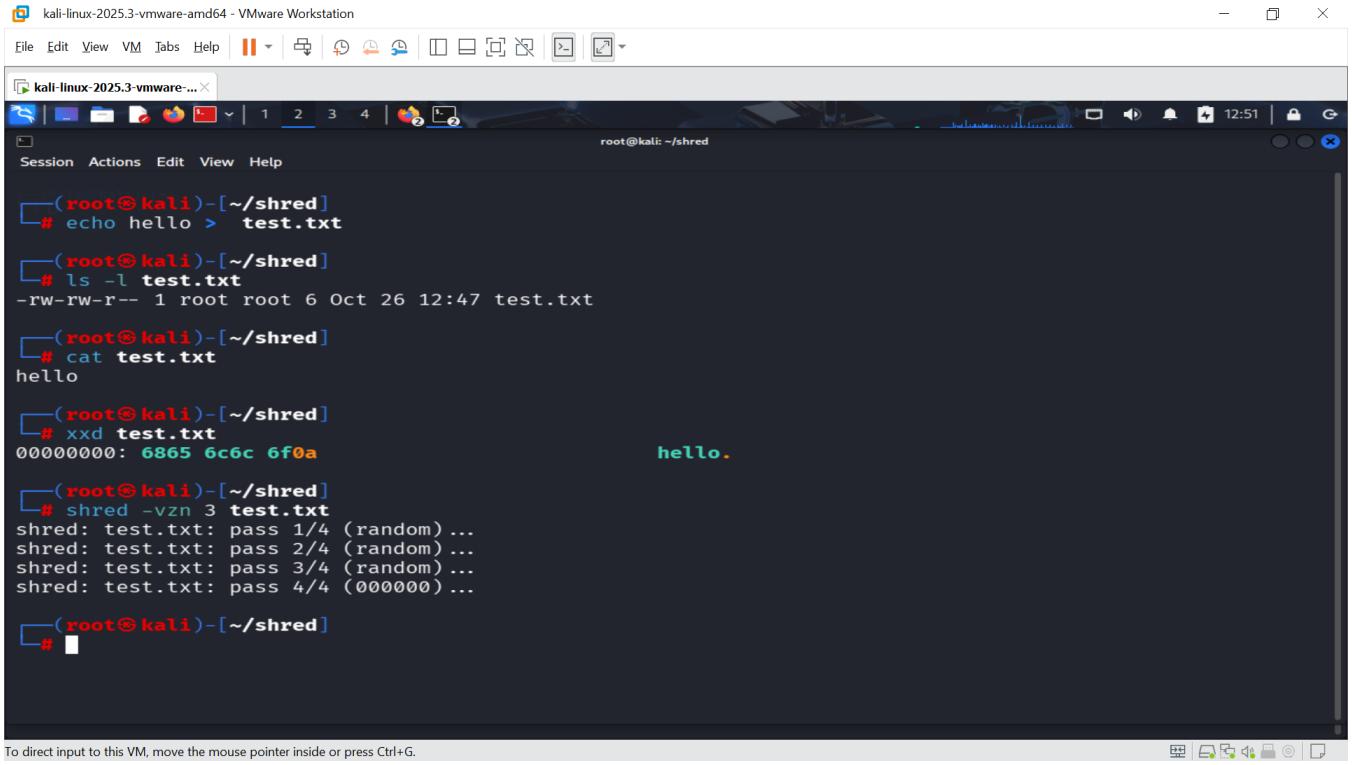
Use only the materials supplied through E-Campus and the accompanying teaching decks. The course sequence includes a USB DD image, an FTP trace (rhino.log), HTTP traces (rhino2.log and rhino3.log), and

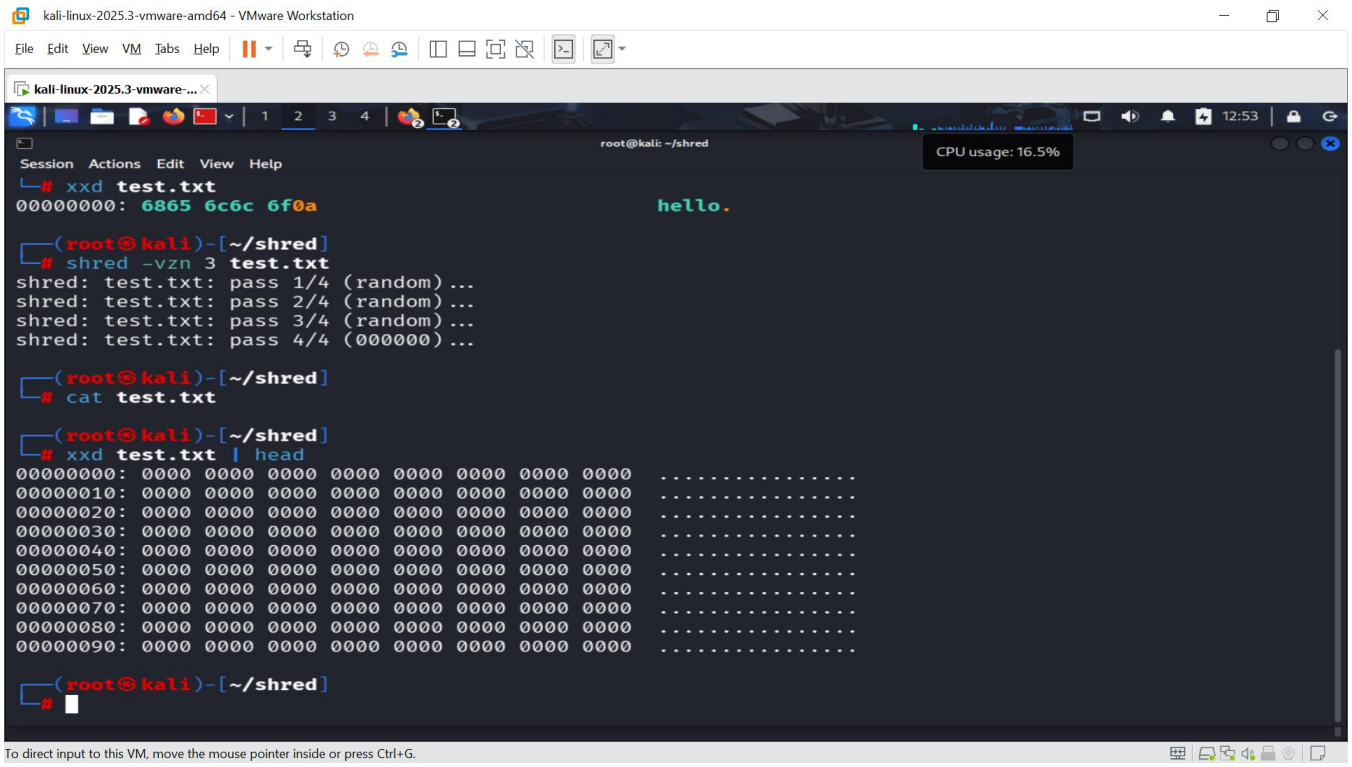
supporting recovery and Wireshark guidance. Use the exact filename in your evidence package if it differs from the deck. Before analysis, create a working copy and record integrity information for every original file. Do not alter, shred, or dispose of original course material before your work has been assessed.

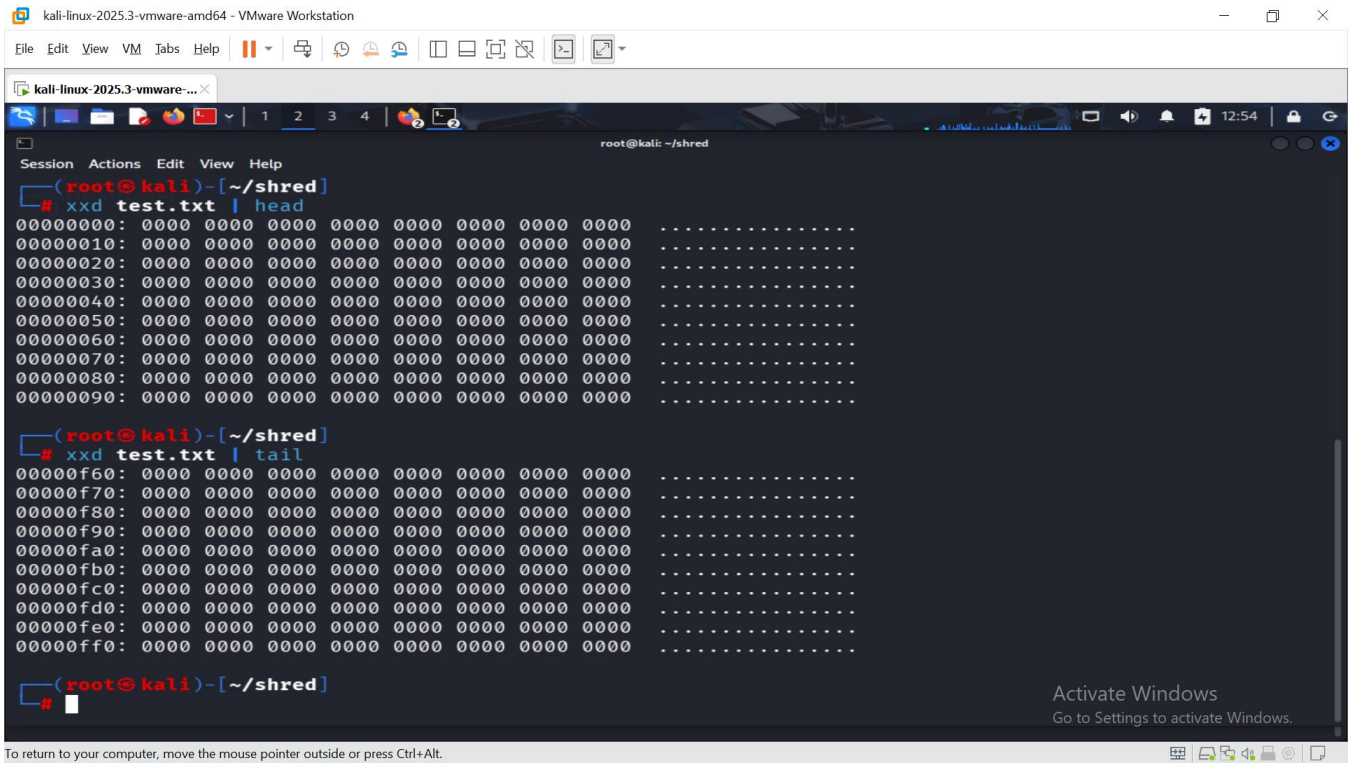
shred --help



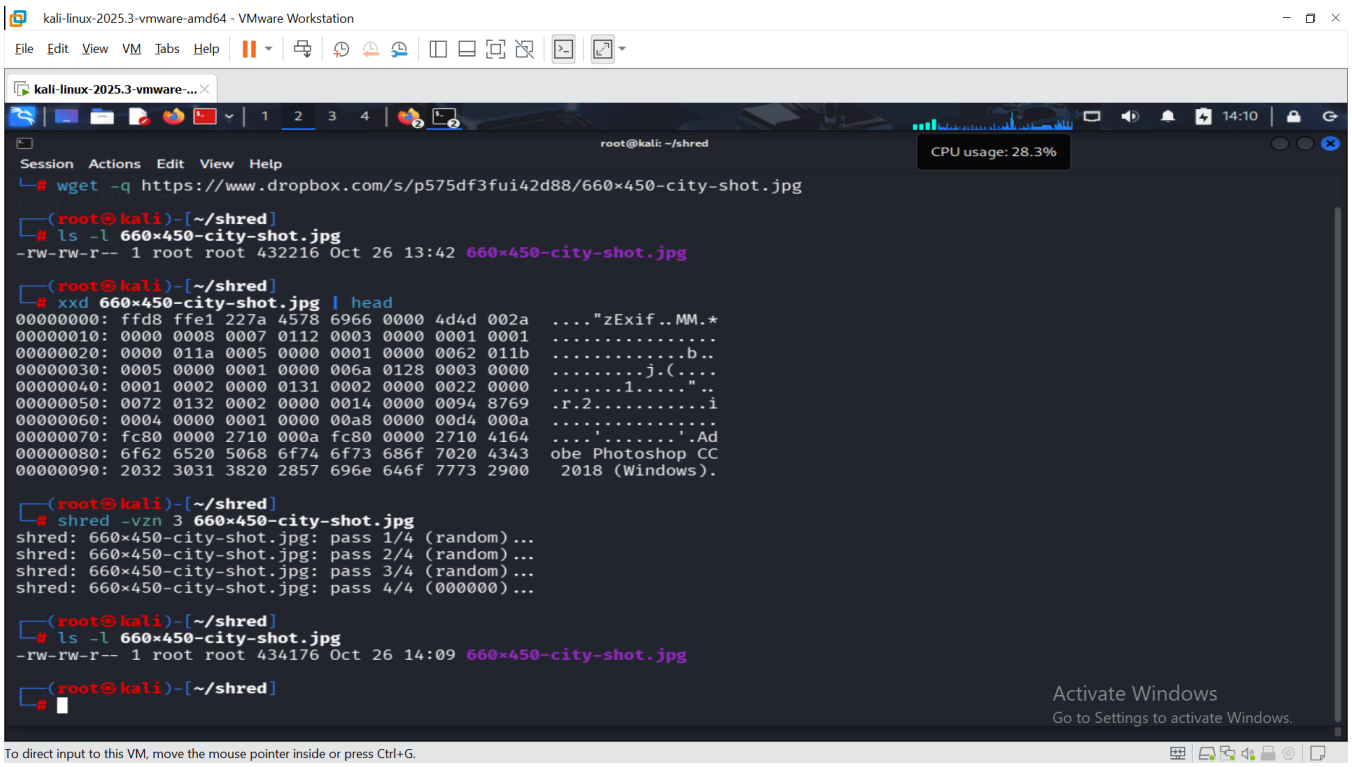



kali-linux-2025.3-vmware-amd64 - VMware Workstation
File Edit View VM Tabs Help
kali-linux-2025.3-vmware-...
12:51
root@kali: ~/shred
Session Actions Edit View Help
┌──(root㉿kali)-[~/shred]
└─# echo hello > test.txt
┌──(root㉿kali)-[~/shred]
└─# ls -l test.txt
-rw-rw-r-- 1 root root 6 Oct 26 12:47 test.txt
┌──(root㉿kali)-[~/shred]
└─# cat test.txt
hello
┌──(root㉿kali)-[~/shred]
└─# xxd test.txt
00000000: 6865 6c6c 6f0a hello.
┌──(root㉿kali)-[~/shred]
└─# shred -vzn 3 test.txt
shred: test.txt: pass 1/4 (random)...
shred: test.txt: pass 2/4 (random)...
shred: test.txt: pass 3/4 (random)...
shred: test.txt: pass 4/4 (000000)...
┌──(root㉿kali)-[~/shred]
└─#
To direct input to this VM, move the mouse pointer inside or press Ctrl+G.


kali-linux-2025.3-vmware-amd64 - VMware Workstation
File Edit View VM Tabs Help
kali-linux-2025.3-vmware-...
12:53
root@kali: ~/shred
CPU usage: 16.5%
Session Actions Edit View Help
└─# xxd test.txt
00000000: 6865 6c6c 6f0a hello.
┌──(root㉿kali)-[~/shred]
└─# shred -vzn 3 test.txt
shred: test.txt: pass 1/4 (random)...
shred: test.txt: pass 2/4 (random)...
shred: test.txt: pass 3/4 (random)...
shred: test.txt: pass 4/4 (000000)...
┌──(root㉿kali)-[~/shred]
└─# cat test.txt
┌──(root㉿kali)-[~/shred]
└─# xxd test.txt | head
00000000: 0000 0000 0000 0000 0000 0000 0000 0000 ....
00000010: 0000 0000 0000 0000 0000 0000 0000 0000 ....
00000020: 0000 0000 0000 0000 0000 0000 0000 0000 ....
00000030: 0000 0000 0000 0000 0000 0000 0000 0000 ....
00000040: 0000 0000 0000 0000 0000 0000 0000 0000 ....
00000050: 0000 0000 0000 0000 0000 0000 0000 0000 ....
00000060: 0000 0000 0000 0000 0000 0000 0000 0000 ....
00000070: 0000 0000 0000 0000 0000 0000 0000 0000 ....
00000080: 0000 0000 0000 0000 0000 0000 0000 0000 ....
00000090: 0000 0000 0000 0000 0000 0000 0000 0000 ....
┌──(root㉿kali)-[~/shred]
└─#
To direct input to this VM, move the mouse pointer inside or press Ctrl+G.



Command used: wget -q https://www.dropbox.com/s/p575df3fui42d88/660x450-city-shot.jpg



Original image: 660x450-city-shot.jpg

-rw-rw-r-- 1 root root 432216 Oct 26 13:42 660x450-city-shot.jpg

-rw-rw-r-- 1 root root 434176 Oct 26 14:09 660x450-city-shot.jpg

wget -q https://raw.githubusercontent.com/frankwxu/digital-forensics-lab/main/Illegal_Possession_Images/USB_image/DFRWS2005-RODEO.zip

Command used: openssl dgst -md5 rhino.log

MD5(rhino.log)= c0d0093eb1664cd7b73f3a5225ae3f30

Command used: openssl dgst -md5 rhino2.log

MD5(rhino2.log)= cd21eaf4acfb50f71ffff857d7968341

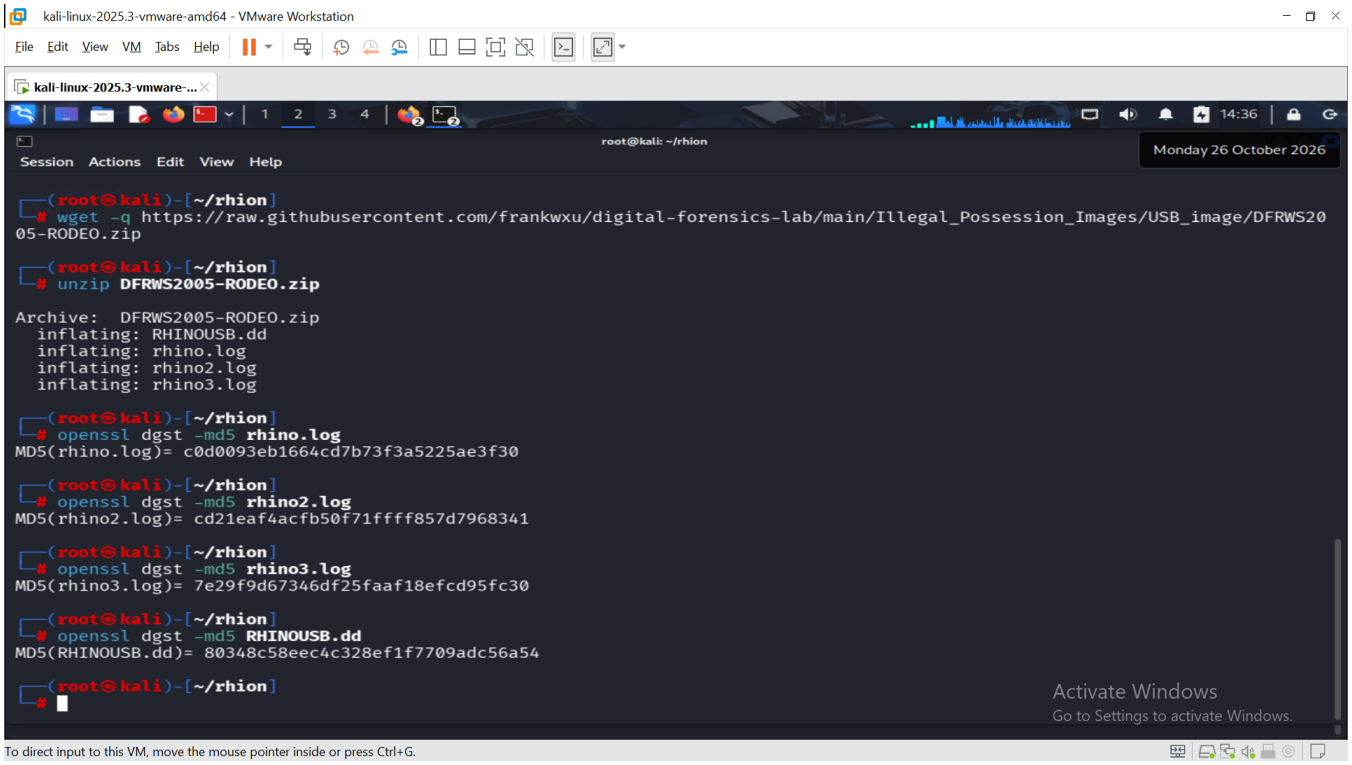
Command used: openssl dgst -md5 rhino3.log

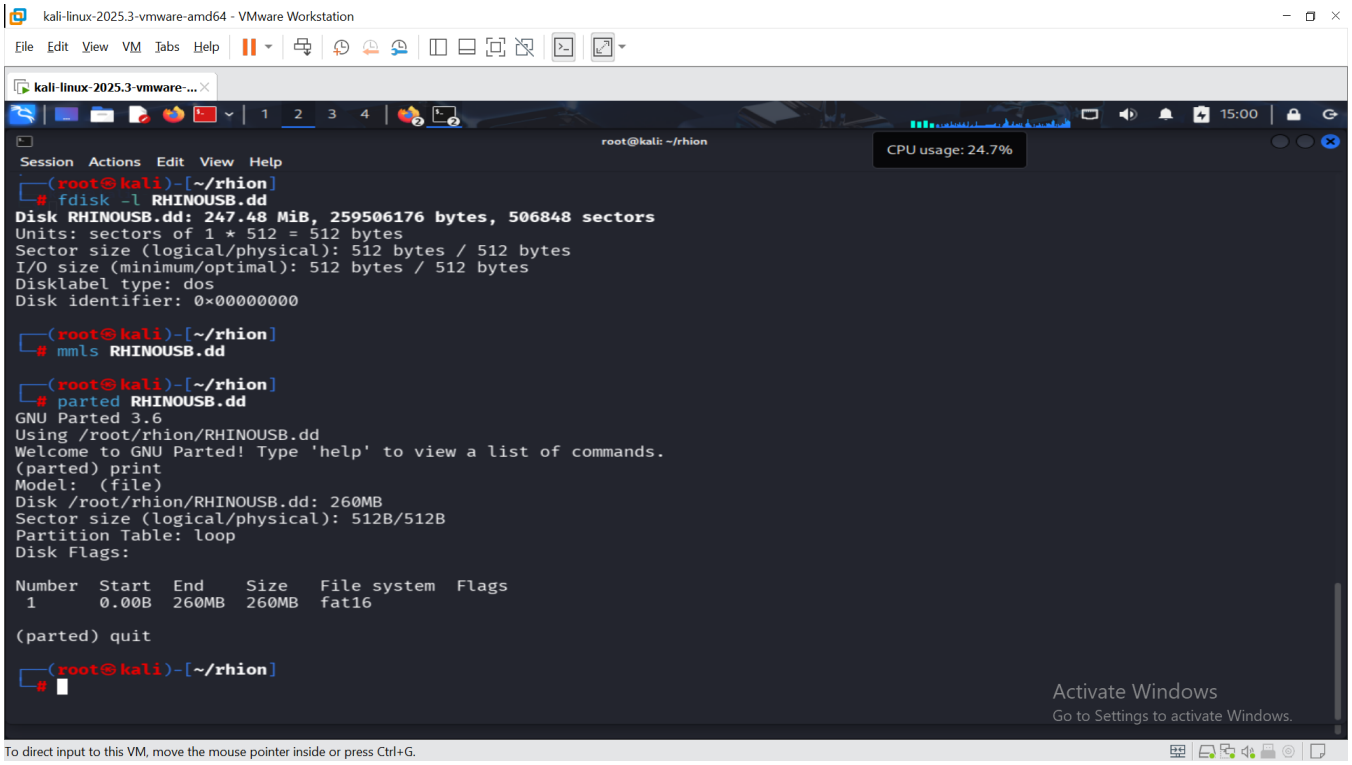
MD5(rhino3.log)= 7e29f9d67346df25faaf18efcd95fc30

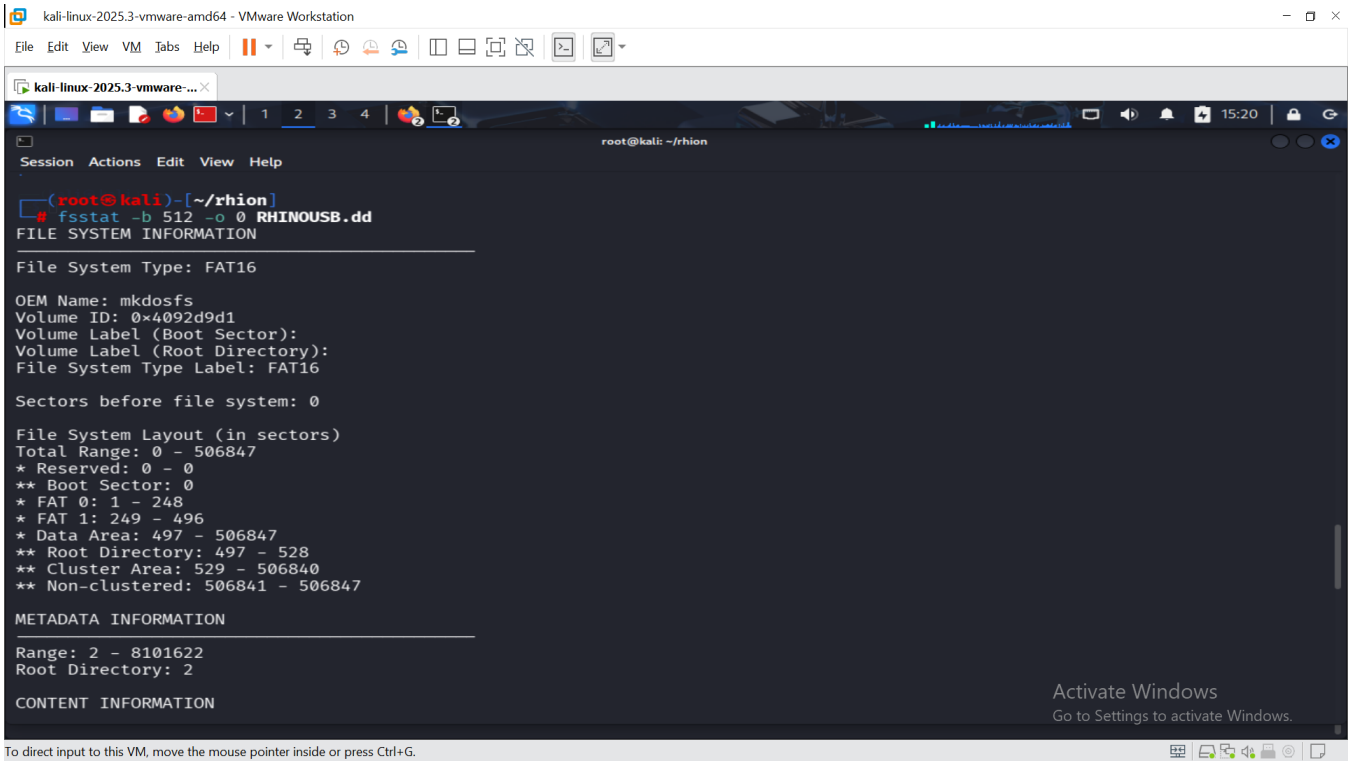
Command used: openssl dgst -md5 RHINOUSB.dd

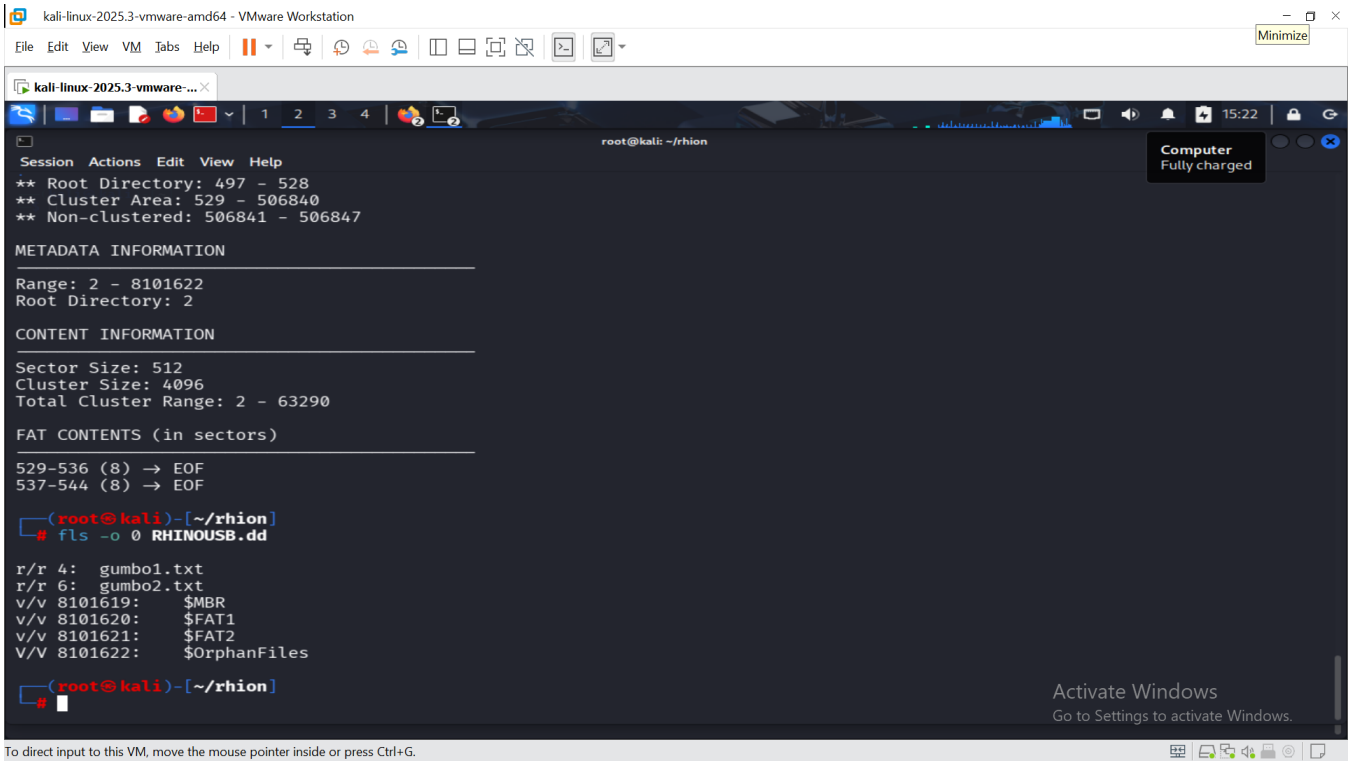
MD5(RHINOUSB.dd)= 80348c58eec4c328ef1f7709adc56a54

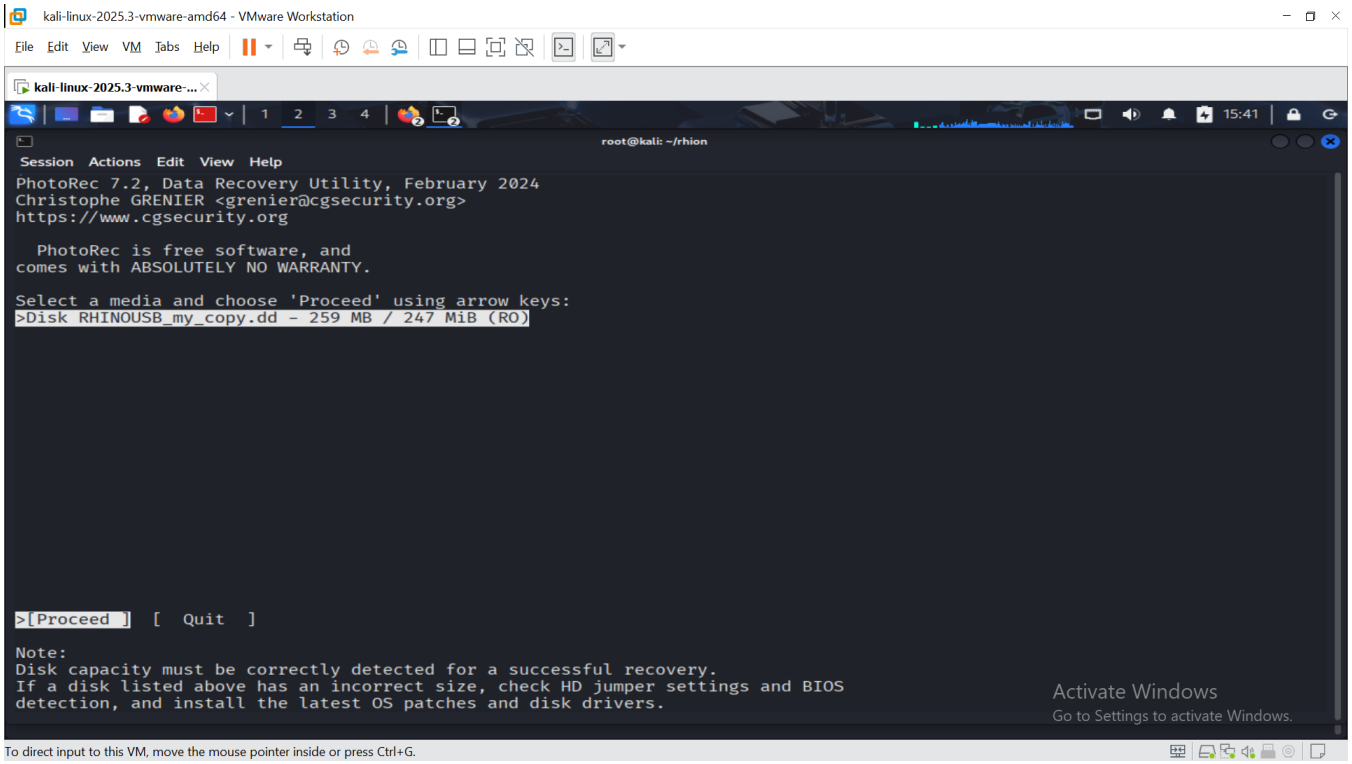
See screenshot bellow



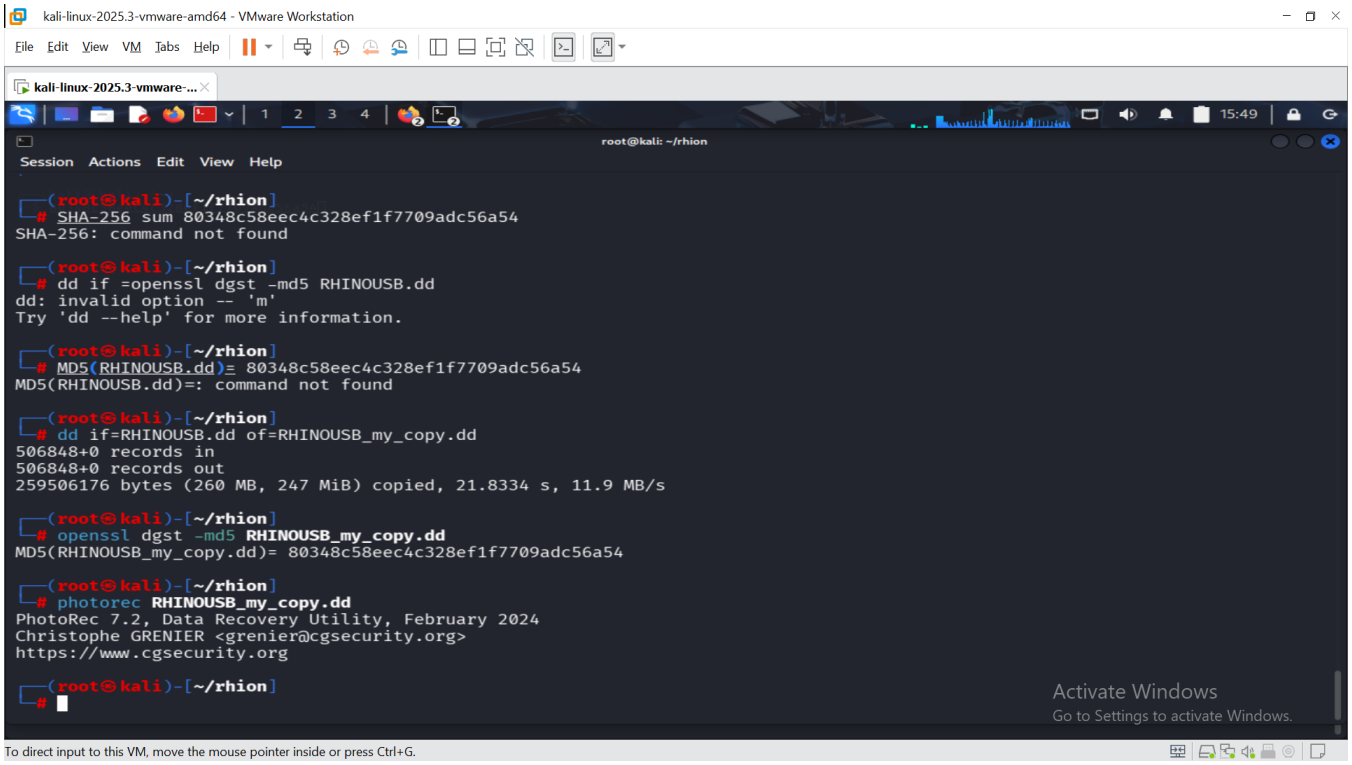

kali-linux-2025.3-vmware-amd64 - VMware Workstation
File Edit View VM Tabs Help
kali-linux-2025.3-vmware-...
15:00
root@kali: ~/rhion
CPU usage: 24.7%
Session Actions Edit View Help
(root㉿kali)-[~/rhion]
fdisk -l RHINOUSB.dd
Disk RHINOUSB.dd: 247.48 MiB, 259506176 bytes, 506848 sectors
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
Disklabel type: dos
Disk identifier: 0x00000000
(root㉿kali)-[~/rhion]
mmls RHINOUSB.dd
(root㉿kali)-[~/rhion]
parted RHINOUSB.dd
GNU Parted 3.6
Using /root/rhion/RHINOUSB.dd
Welcome to GNU Parted! Type 'help' to view a list of commands.
(parted) print
Model: (file)
Disk /root/rhion/RHINOUSB.dd: 260MB
Sector size (logical/physical): 512B/512B
Partition Table: loop
Disk Flags:
Number Start End Size File system Flags
1 0.00B 260MB 260MB fat16
(parted) quit
(root㉿kali)-[~/rhion]
#
Activate Windows
Go to Settings to activate Windows.
To direct input to this VM, move the mouse pointer inside or press Ctrl+G.


kali-linux-2025.3-vmware-amd64 - VMware Workstation
File Edit View VM Tabs Help
kali-linux-2025.3-vmware-...
15:20
root@kali: ~/rhion
Session Actions Edit View Help
(root㉿kali)-[~/rhion]
fsstat -b 512 -o 0 RHINOUSB.dd
FILE SYSTEM INFORMATION
File System Type: FAT16
OEM Name: mkdosfs
Volume ID: 0x4092d9d1
Volume Label (Boot Sector):
Volume Label (Root Directory):
File System Type Label: FAT16
Sectors before file system: 0
File System Layout (in sectors)
Total Range: 0 - 506847
* Reserved: 0 - 0
** Boot Sector: 0
* FAT 0: 1 - 248
* FAT 1: 249 - 496
* Data Area: 497 - 506847
** Root Directory: 497 - 528
** Cluster Area: 529 - 506840
** Non-clustered: 506841 - 506847
METADATA INFORMATION
Range: 2 - 8101622
Root Directory: 2
CONTENT INFORMATION
Activate Windows
Go to Settings to activate Windows.
To direct input to this VM, move the mouse pointer inside or press Ctrl+G.





Command used: photorec RHINOUSB_my_copy.dd



What you must submit

Item	Required output	What to provide
1	Forensic working set	Record the supplied disk image: packet captures, and any recovered artefact by original file name, byte size, SHA-256 value, source location, and working-copy location. Preserve the original course material unchanged.
2	Recovery and analysis record	Keep labelled output or screenshots for the disk-image layout, file recovery, metadata review, steganography testing, FTP and HTTP analysis, reconstructed objects, and validation checks.
3	Correlated timeline	Create a UTC, time-ordered account with at least five material events across at least three artefact families. Cite packet numbers or streams, file paths, hashes, timestamps, and evidence-log IDs.
4	Student report	Submit one report in PDF or DOCX format with a completed evidence log and timeline appendix. Include only the supporting files requested by your lecturer.

Investigation task

Conduct the examination in stages: preserve the supplied material; recover and validate relevant files from the disk image; examine media for hidden content; reconstruct and validate material FTP and HTTP objects; then correlate the artefacts in a single UTC timeline. Base every finding on identifiable records such as a source filename, hash, partition or metadata record, packet number, TCP stream, timestamp, response header, tool output, or screenshot. The questions below define the minimum scope of the report.

No	Investigation question	Minimum evidence to discuss
1	How did you preserve the case materials?	For each supplied disk image: 660x450-city-shot.jpg and capture, record source, original file name: 660x450-city-shot.jpg byte size: 512 , SHA-256 value, working-copy method, tool version, and file-access precautions.
2	What does the USB disk image show before recovery?	Document the partition layout, file-system information, allocated or unallocated area examined, relevant directory or metadata records, and any limits of file-system metadata.
3	What deleted or recoverable files are supported?	Describe the recovery method, output folder, recovered-file type, hash, visual or metadata check, and whether files are unique or duplicates. Do not report deck answers as your own findings.
4	Is there evidence of hidden content in the recovered media?	Record metadata review, steganography detection results, the authorised offline recovery method used, recovered output, hashes, and the evidence that supports or weakens a hiding interpretation.
5	What does the FTP capture establish?	Identify relevant sessions, transfer direction, packet or stream references, server response, object boundaries, reconstructed file validation, and timestamps. Treat any visible credential as sensitive evidence; do not reuse it.
6	What does the protected archive analysis establish?	Show that the reconstructed archive is complete, identify the evidence of encryption, and document any authorised course-only password-recovery workflow. Compare recovered content with other evidence using hashes where available.
7	What does the HTTP capture establish?	Identify material requests and responses, client and server endpoints, content type, stream or packet references, object extraction and validation. Do not execute any reconstructed executable or contact a recorded host.
8	What conclusion can you defend?	Build a UTC timeline, distinguish direct observations from inference, state a confidence level, and explain at least three limitations or alternative explanations concerning possession, transfer, and user attribution.

Technical starting points

The following are course-aligned starting points, not a substitute for documenting your own method. Work only on copies. Replace placeholders with values from your evidence and preserve the outputs that substantiate each decision.

Purpose	Course-aligned starting point
Preserve and identify evidence	Create read-only originals and working copies. Record SHA-256 values before and after copying; use file type and hex inspection only on working copies.
Examine the disk image	Use the Sleuth Kit workflow demonstrated in class to review media layout and file-system records (for example, mmls, fsstat, fls, and icat where appropriate).
Recover deleted files	Use the authorised PhotoRec or file-carving workflow on the working copy. Record the recovery scope selected and the output location; hash every material recovered file.

Purpose	Course-aligned starting point
Review media and hidden data	Review file type and metadata, then use the supplied steganography workflow to detect and, if justified, recover embedded course content. Keep detector output and recovery notes.
Review FTP traffic	Use Wireshark display filtering and Follow TCP Stream to locate control and data sessions. Preserve packet or stream numbers, direction, timestamps, and raw object boundaries.
Review HTTP traffic	Use http, http.request, http.response, and tcp.stream eq <number> as starting filters. Confirm reassembly, response headers, content type, and file boundaries before exporting a working copy.
Validate reconstructed objects	Use magic number and trailer checks, file-type output, byte size, and SHA-256 hashes. A matching hash can establish binary equality but does not identify a person or original source by itself.

Evidence interpretation safeguards

- Use a hash and read-only preservation record for each original. A recovered working copy or carved file must remain traceable to its source and recovery method.
- For network objects, preserve the relevant packet range or TCP-stream reference, transfer direction, and reassembly method. A single packet is not necessarily a complete file.
- Visible FTP or email credentials are sensitive evidence. Record only what is necessary for assessment, redact them in screenshots where appropriate, and never reuse them outside the supplied course artefacts.
- An authorised password-recovery step may be used only against the supplied training archive and only within the course lab. Do not apply recovered passwords, wordlists, or methods to any other file, account, host, or service.
- A file hash, IP address, host name, username, or directory record can corroborate a relationship, but it does not alone prove original source, completion of an action, physical possession, or human identity.
- Do not execute a reconstructed executable, connect to a recorded endpoint, submit material to public scanners, or upload case evidence outside the approved lab. Follow lecturer-approved secure-disposal instructions only after assessment.

Required report structure

1. Case reference and student identification. Include your full name, registration number, course code, and case title.
2. Executive conclusion. State a concise, cautious conclusion and confidence level. Do not present a person-level attribution unless the artefacts support it.
3. Evidence acquisition and integrity. Record sources, filenames, sizes, hashes, read-only originals, working copies, tools, versions, and access controls.
4. Method. Explain the disk-image, recovery, steganography, FTP, HTTP, Wireshark, object-validation, and time-normalisation methods used.
5. Findings. Answer every investigation question with a precise evidence reference: file path, hash, metadata record, packet number, stream, timestamp, header, or screenshot ID.
6. Timeline and correlation. Present the material sequence in UTC. State how artefacts corroborate, conflict with, or fail to establish each proposed relationship.
7. Conclusion and limitations. Explain gaps, alternative explanations, duplicate files, partial captures, encryption, timestamp uncertainty, and the distinction between technical account evidence and a person at a keyboard.
8. Appendices. Include labelled screenshots, selected tool output, the completed timeline, the completed evidence log, and only the supporting material requested by your lecturer.

Timeline template

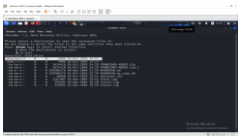
Complete a version of this table in your report or evidence appendix. Add rows for all material events and keep the main timeline in UTC.

Stage	UTC timestamp	Evidence ID	Source artefact	Observed fact	Interpretation and limit
Media preservation			660x450-city-shot.jpg	Change in the size of the original copy and working copy from 432216 To 434176	
Disk image / file-system review					
Recovered or hidden content					
FTP transfer activity					
HTTP transfer activity					
Corroborated conclusion					

Evidence log template

Use an evidence log that lets an assessor locate the exact source file, tool, command or workflow, packet or stream, record, hash, and screenshot that supports each finding.

Evidence ID	Tool and method	Source record, stream, or path	Observed fact	Screenshot or appendix reference
E01	Command used: openssl dgst -md5 rhino.log MD5(rhino.log)= c0d0093eb1664cd7b73f3a5225ae3f30			
E02	Command used: openssl dgst -md5 rhino3.log MD5(rhino3.log) = 7e29f9d67346df25faaf18efcd95fc30			

Evidence ID	Tool and method	Source record, stream, or path	Observed fact	Screenshot or appendix reference
E03	Command used: openssl dgst -md5 rhino3.log MD5(rhino3.log) = 7e29f9d67346df25faaf18efcd95fc30			
E04	Command used: openssl dgst -md5 RHINOUSB.dd MD5(RHINOUSB.dd)= 80348c58eec4c328ef1f7709adc56a54			
E05	openssl dgst -md5 RHINOUSB.dd MD5(RHINOUSB.dd)= 80348c58eec4c328ef1f7709adc56a54			
E06				
E07				

Submission rules

- Work individually. Your analysis, report wording, screenshots, timeline, and evidence log must be your own work.
- Submit the report as SBTDF204_CaseStudy4_RegistrationNumber.pdf or .docx. If separate supporting material is requested, use SBTDF204_CaseStudy4_Evidence_RegistrationNumber.zip.
- Do not upload the original disk image, packet capture, recovered archive, media file, or executable unless your lecturer specifically requests it. Submit only the report and required supporting evidence.
- Use only the authorised, offline course artefacts in the approved lab environment. Do not contact people, accounts, hosts, or services; do not reuse credentials; do not execute recovered binaries; and do not publicly upload or share case evidence.
- Redact unrelated personal or sensitive data in screenshots while retaining enough packet, stream, path, timestamp, hash, and tool detail for the examiner to verify your finding.

Marking guide

The E-Campus maximum score applies to this case study. The assessor will apply the following percentage weighting to that maximum score.

Criterion	What strong work demonstrates	Weight
Evidence acquisition and recovery	Preserves the supplied material, documents integrity and working copies, and uses a defensible disk-image and deleted-file recovery workflow.	15%
Recovered-file and steganography analysis	Validates recovered media, records metadata and steganography results, and supports findings with clear tool output, hashes, and limitations.	20%
FTP and protected-archive analysis	Analyses FTP sessions, reconstructs and validates material objects, and documents the authorised course-only archive workflow without reusing credentials.	20%
HTTP and Wireshark traffic analysis	Uses request, response, stream, and packet evidence to extract and validate relevant objects while accounting for reassembly and transfer direction.	20%
Correlation, timeline, conclusion, and limitations	Builds a coherent UTC timeline, distinguishes observations from inferences, corroborates sources, and gives a cautious conclusion.	15%
Report quality and evidence presentation	Writes clearly, labels screenshots and output, cites evidence IDs consistently, and supplies a usable evidence log.	10%

Final check before submission

- Every major conclusion cites a source artefact, method, hash or packet/stream reference, timestamp, and screenshot or appendix reference.
- Your timeline has at least five material events across at least three artefact families and uses UTC consistently.
- Your report explains how deleted-file recovery, hidden-content analysis, FTP, and HTTP evidence corroborate or challenge the proposed sequence.
- Your conclusion distinguishes technical evidence of a file, transfer, account, or endpoint from proof of possession, transfer completion, and a named person.